Two satellites

X22 (2022) — English translation

In 1973, a method was proposed for launching satellites into hyperbolic orbit using a gravitational maneuver, which you are invited to analyze. Within the framework of this problem, the motion of two satellites 1 and 2, launched from opposite poles of the Earth with the same magnitude and direction velocities v tangential to it, is studied. Satellite 2 is launched into orbit a short time τ after the launch of the first. If only one of the satellites is launched into orbit, its trajectory will be a parabola. The mass of satellite 1 is M_1, and satellite 2 is $M_2 \gg M_1$. The resistance of gas in the atmosphere can be neglected, as well as the influence of gravitational attraction from other bodies. The mass of the Earth is $M_3 \gg M_2$, the radius of the Earth is R_3, the gravitational constant is G. Let us denote by P the point of intersection of the elliptical orbits of the satellites, and by φ the angle between the directions of their velocities at this point. The movement of satellite 1 consists of three sections: 1) Movement in a parabolic orbit; 2) Interaction with satellite 2 near point P; 3) Further movement along a hyperbolic orbit in the Earth's gravity field.

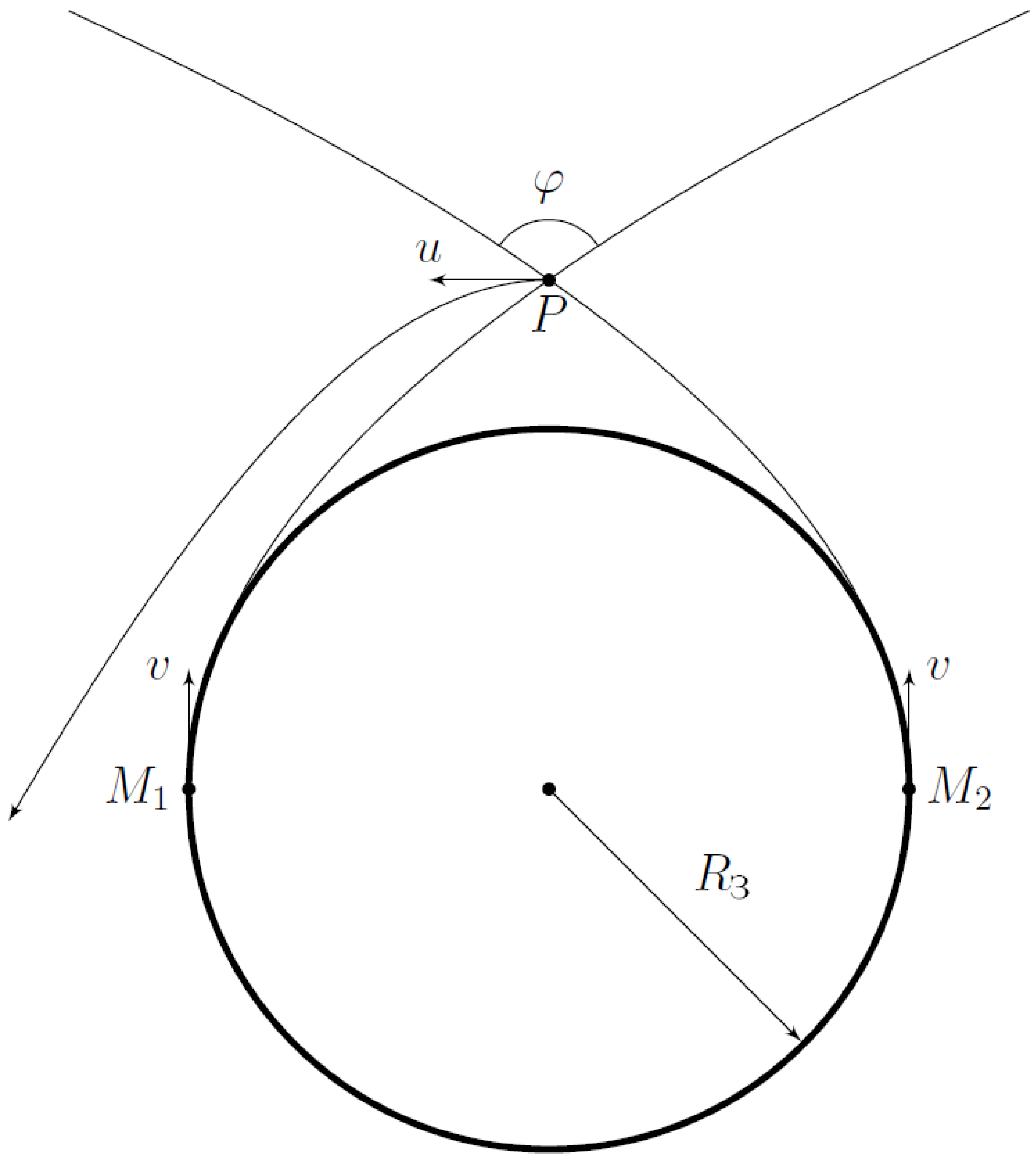

Fig. 1.

Part A. Trajectory of one of the satellites (3.3 points)

In this part of the problem, you will have to approximately describe the trajectory of the satellites to the intersection point P of their orbits.

A1	What is the speed of the satellites v at the moment of launch? Express your answer in terms of G, M_3 and R_3.	**0.40**

A2	At what distance from the center of the Earth R_0 is the point P of intersection of the satellite trajectories? Express your answer using R_3.	**0.60**

A3	What is the angle φ between the directions of satellite velocities at point P?	**0.80**

A4	Find the time t_0 after which the satellites reach point P. Express your answer using G, R_3 and M_3.	**1.50**

Hereinafter, assume that $\tau \ll t_0$. When satellites approach point P, their interaction with the Earth can be neglected.

Part B. Satellite Interaction (4.0 points)

B1	Assuming that near the point P the satellites moved along straight lines, the directions of which coincided with the directions of their velocities at this point, find the relative speed of the satellites v_0, as well as the minimum distance b between them, neglecting their gravitational interaction. Express your answer using v and τ.	**0.70**

Next, consider that in the reference frame of satellite 2, the orbit of satellite 1 is a hyperbola with an impact parameter b at their infinite distance. Also consider that at an infinite distance the relative speed of the satellites is equal to v_0 found in the previous paragraph.

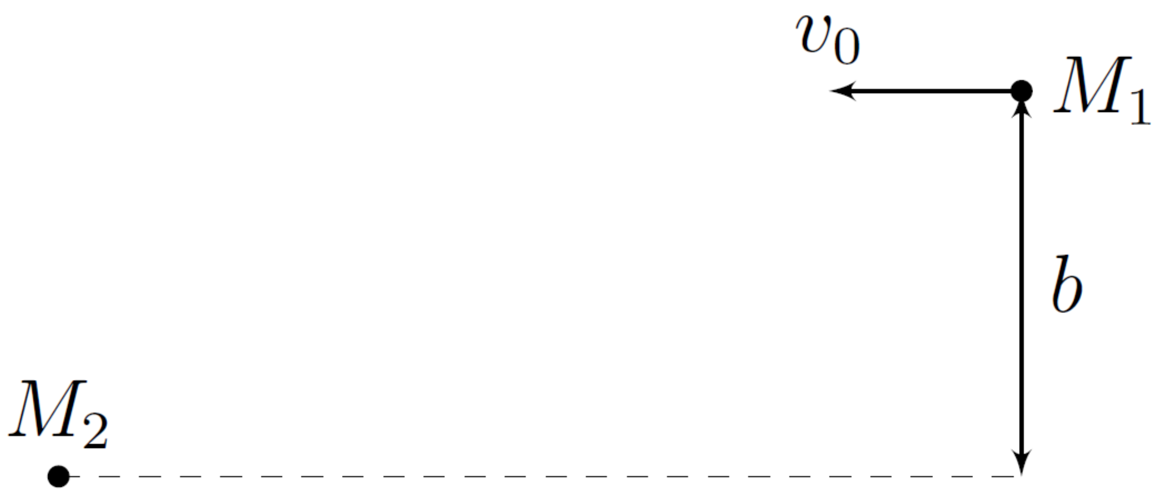

Fig. 2.

When the interaction of satellites 1 and 2 can again be neglected, the speed of satellite 1 in the Earth's reference frame is equal to u and is directed perpendicular to the line connecting it to the Earth.

B2	Find u. Express your answer using v.	**1.00**

B3	Find τ. Express your answer in terms of G, M_2, M_3 and R_3.	**2.30**

Part C. Repeated motion in the Earth's gravity field (2.7 points)

After interaction with satellite 2, the orbit of satellite 1 in the Earth's gravity field is a hyperbola.

C1	Find the speed v_∞ of satellite 1 at an infinite distance from the Earth. Express your answer using v.	**1.00**
C2	Find the angle φ_∞ between vectors $\vec{u}$ and $\vec{v}_\infty$.	**1.50**
C3	Is it really possible to put a satellite into orbit this way? Justify your answer.	**0.20**

Source: https://pho.rs/p/277